Randy Roberts

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Skills

Languages: C#, C++, Java, Python, JavaScript, TypeScript, HTML, CSS, Fortran, Lua
Frameworks: Unity, React, Next.js, Qt, WPF, WinForms, Retool
Tools: Windows, Linux, WSL, Docker, Visual Studio, CMake, Git, JIRA, Confluence, Jenkins

Experience

Flight Simulation Software Engineer, Boeing – St. Louis, MO Aug 2025 – Present

- Diagnose and resolve input defects on simulated aircraft displays, so pilot interactions register reliably during customer training.
- Refactor the aircraft display data pipeline so each display runs at its own configurable refresh rate, decoupling systems that previously shared a single update cycle.
- Deliver quality-of-life improvements to the team's internal process manager, including showing which machine each process runs on and opening an executable's file location in one click, making it faster to locate and manage hundreds of simulation processes across dozens of machines.
- Track down and fix defects in the core simulation logic across C++, C#, and Fortran subsystems, documenting root causes so the team can diagnose similar issues faster.

Flight Simulation Software Engineer, AVT Simulation – Orlando, FL Jun 2024 – Jun 2025

- Architected and built a C++ radio communications microservice, shipped to a customer, that relays traffic among dozens of simulated radios across a distributed simulator over UDP.
- Implemented a UDP interface between the radio client and the simulation, defining the message protocol that keeps radio state and transmissions synchronized in real time during live training.
- Designed a runtime process manager in Java, delivered to a customer through Agile sprint cycles, that launches and terminates simulation executables on demand, automating process control during live runs.
- Added structured logging to the after-action review tools, giving developers clearer visibility into system behavior and speeding up debugging of training sessions.

Software Engineer Intern, Alidade Systems – Remote Mar 2024 – Jun 2024

- Designed and prototyped the company's financial forecasting and budgeting user interface in Retool and JavaScript across frontend and backend, partnering with engineers and product leads to deliver it to the company's first clients.
- Enabled budgets to scale in both breadth and depth by designing data structures for hierarchical financial data. This work was featured in product demos to prospective business partners.

Web Developer Intern, Arising.co – Orlando, FL Jul 2023 – Dec 2023

- Shipped three production Next.js websites for Arising's clients and partners, owning each from build through launch and ongoing maintenance against client requirements and deadlines.
- Raised the performance and search visibility of client sites by establishing consistent frontend optimization practices across projects, so every launch met the same quality bar.

Projects

Unity Movement Framework github.com/RandyRoberts99/UnityMovementFramework

- Build a reusable foundation for character movement in video games, so new moves like dashing or wall running can be added as plug-ins without rewriting existing code.
- Design a simple framework for other developers to pick up and extend, with clear organization and documentation, and controls that feel smooth and responsive on any hardware.

Education

University of Central Florida – BS in Computer Science May 2024